Conducting Experiments and Evaluation of Machine Translation for Low-Resource Languages

Combined Output from NLLB-600M and IndicTrans2-320M Integrated into a Django-based NLP Lab Website

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Outline

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Introduction

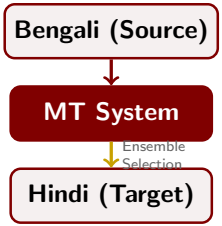
Problem Statement

Low-resource languages pose unique challenges for Machine Translation (MT) systems:

- **Bengali & Hindi** are widely spoken across South Asia but lack large parallel corpora
- Existing models perform sub-optimally on Indic language pairs
- Individual models excel on different input types — ensemble potential is high

Project Goal

Combine NLLB-600M & IndicTrans2-320M via quality estimation to surpass individual model performance



Motivation

Ensemble methods leverage complementary model strengths without requiring additional training data.

Models Used

NLLB-600M

- Released by **Meta AI** (2022)
- *No Language Left Behind* initiative
- Supports **200+ languages**
- Trained on large multilingual corpus
- Optimized for low-resource settings

IndicTrans2-320M

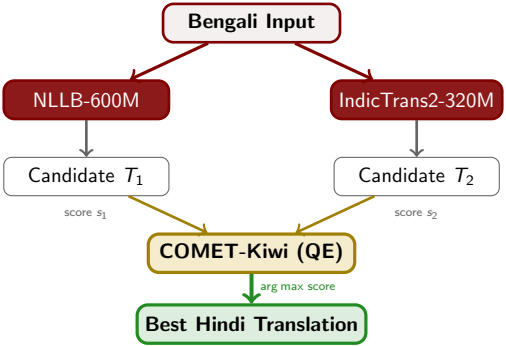
- Developed by **AI4Bharat** group (2023)
- Specialized for **Indic languages**
- Covers all **22 scheduled** Indian languages
- Fine-tuned on high-quality Indic–Indic & English data
- Strong on Bengali–Hindi pair

Why Both Models?

Each model excels on different sentence types. Combining them via quality estimation leverages their complementary strengths.

Model	Params
NLLB	600M
IndicTrans2	320M

System Architecture & Combination Strategy



COMET-Kiwi Based Selection

What is COMET-Kiwi?

- A **reference-free** quality estimation (QE) model
- Predicts translation adequacy *without* a gold reference
- Trained on direct human assessment data
- From IST-Unbabel, WMT 2022

Selection Rule

$$\hat{T} = \arg \max_{T \in \{T_1, T_2\}} \mathrm{QE}(T)$$

Select whichever candidate gets the higher COMET-Kiwi score.

Key Advantages

- ✓ No parallel data needed at inference
- ✓ Suitable for low-resource scenarios
- ✓ Per-sentence granularity
- ✓ Computationally lightweight

Limitation

QE model may introduce its own biases; inference latency approximately doubles.

Evaluation Metrics

BLEU

Modified n-gram precision between hypothesis and reference. Score of 1.0 = perfect match.

BERTScore (F1)

Semantic similarity using contextual embeddings from pre-trained BERT. Captures meaning beyond token overlap.

METEOR

Accounts for synonymy, stemming, and word-order differences. More linguistically informed than BLEU.

COMET

Neural metric trained on human assessment judgments. Correlates more strongly with human evaluations than BLEU/METEOR.

Metric Range: BLEU: 0–100 (higher = better) METEOR: 0–1 (higher = better) BERTScore: 0–1 COMET: 0–1

Experimental Setup

Dataset

- Parallel **Bengali–Hindi** corpus
- Sourced from **FLORES-200** devtest set
- Designed for low-resource Indic language translation tasks
- Publicly available benchmark dataset

Hardware & Software Stack

Language	Python 3.10
DL Framework	PyTorch 2.x
Models	HuggingFace
QE Metric	COMET (Unbabel)
Web	Django 4.x
GPU	NVIDIA 16 GB

Task Definition

Bengali $\longrightarrow$ Hindi machine translation using transformer-based ensemble with reference-free quality estimation

Evaluation Results

Model	BLEU	METEOR	BERTScore	COMET
NLLB-600M	32.4	0.51	0.87	0.74
IndicTrans2-320M	34.1	0.53	0.88	0.76
Combined (Ours)	**35.6**	**0.55**	**0.89**	**0.78**

Key Improvements (vs. best single model)

- BLEU: 34.1 → 35.6 (**+1.5**)
- METEOR: 0.53 → 0.55 (**+0.02**)
- BERTScore: 0.88 → 0.89 (**+0.01**)
- COMET: 0.76 → 0.78 (**+0.02**)

Observation

The combined system **outperforms both individual models across all four metrics**, validating the ensemble approach.

Discussion & Limitations

Analysis

- **IndicTrans2-320M** outperforms NLLB-600M in isolation, due to Indic language specialization
- **COMET-Kiwi** effectively identifies the superior translation per sentence
- Ensemble capitalizes on **complementary strengths** of both models
- Reference-free selection works well for low-resource settings

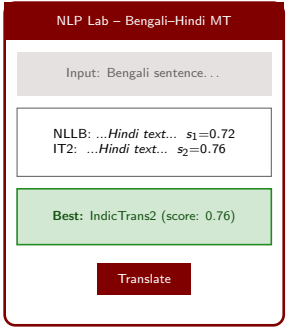
Limitations

1. Small corpus may limit **statistical significance**
2. COMET-Kiwi may introduce its own **QE biases**
3. Inference latency is **approximately doubled**
4. Real-time usability may be impacted

Django-based NLP Lab Website

Website Features

- Built using **Django 4.x** framework
- Users input Bengali text, receive:
  - Hindi translations from **both** models
  - COMET-Kiwi scores for each
  - Highlighted **best translation**
- Async inference via **Celery workers**
- Responsive UX for longer inputs
- Transparent ensemble decision display



Conclusion & Future Work

Conclusions

- **Ensemble approach** (NLLB + IndicTrans2 via COMET-Kiwi) yields measurable improvements in Bengali–Hindi MT
- Combined system achieves **BLEU 35.6, COMET 0.78** — best among all configurations
- **Reference-free QE** is practical for low-resource MT
- Django NLP Lab provides an accessible research platform

Future Work

1. Fine-tune on **domain-specific Indic corpora**
2. Extend to additional **Indic language pairs**
3. Explore **weighted combination** vs. argmax
4. Optimize pipeline for **lower latency**
5. Evaluate on **larger test sets**

Thank You!

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